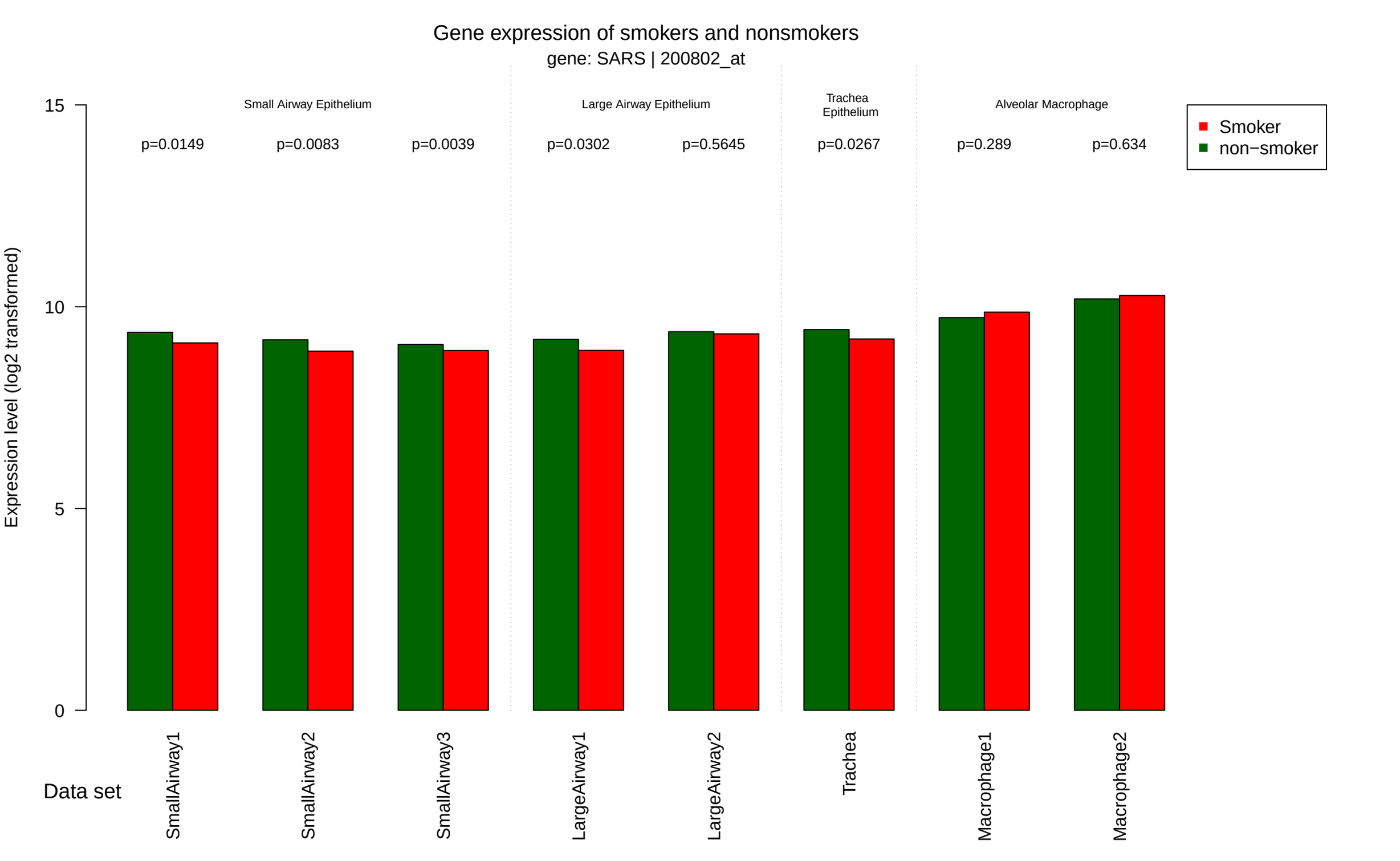
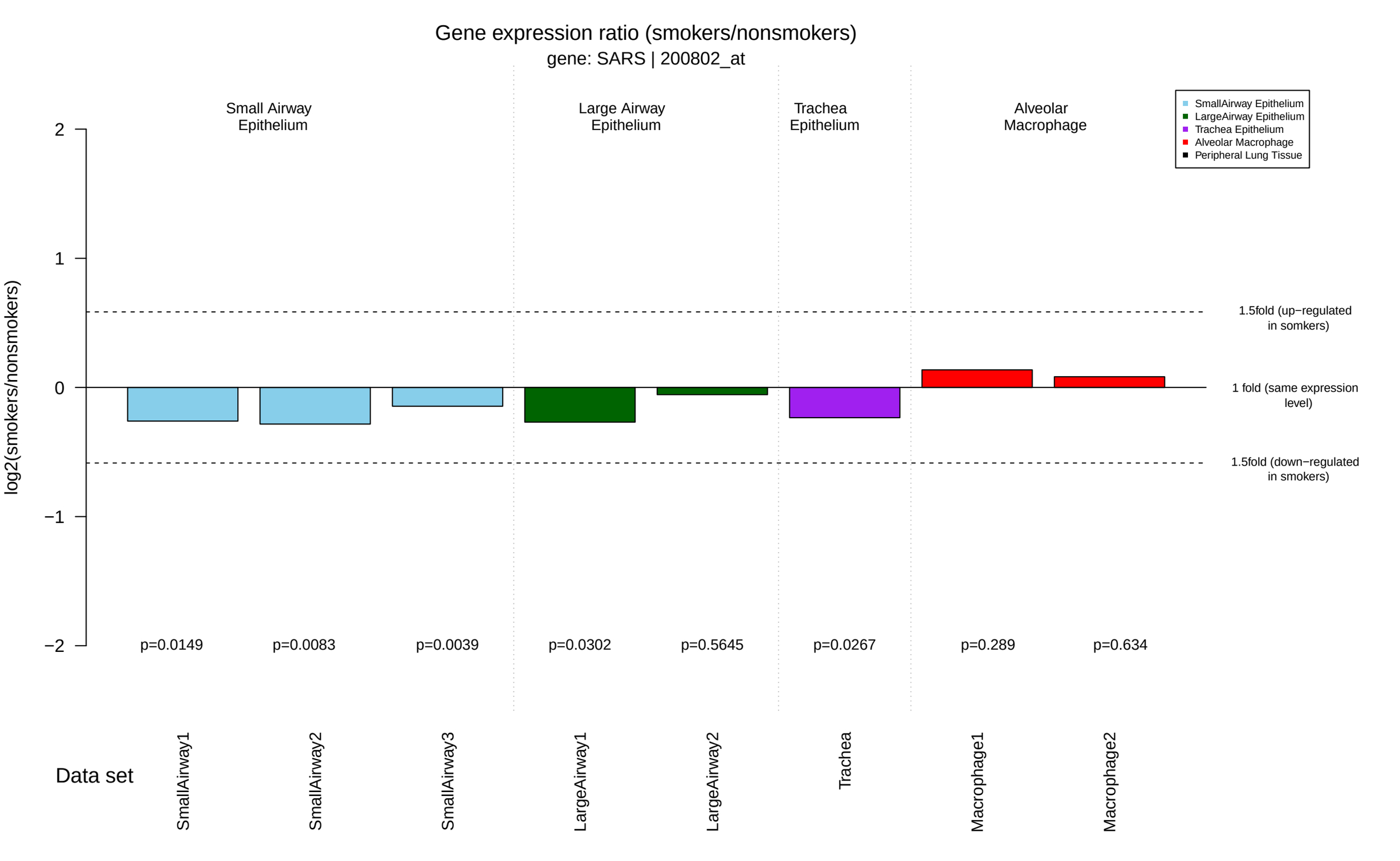
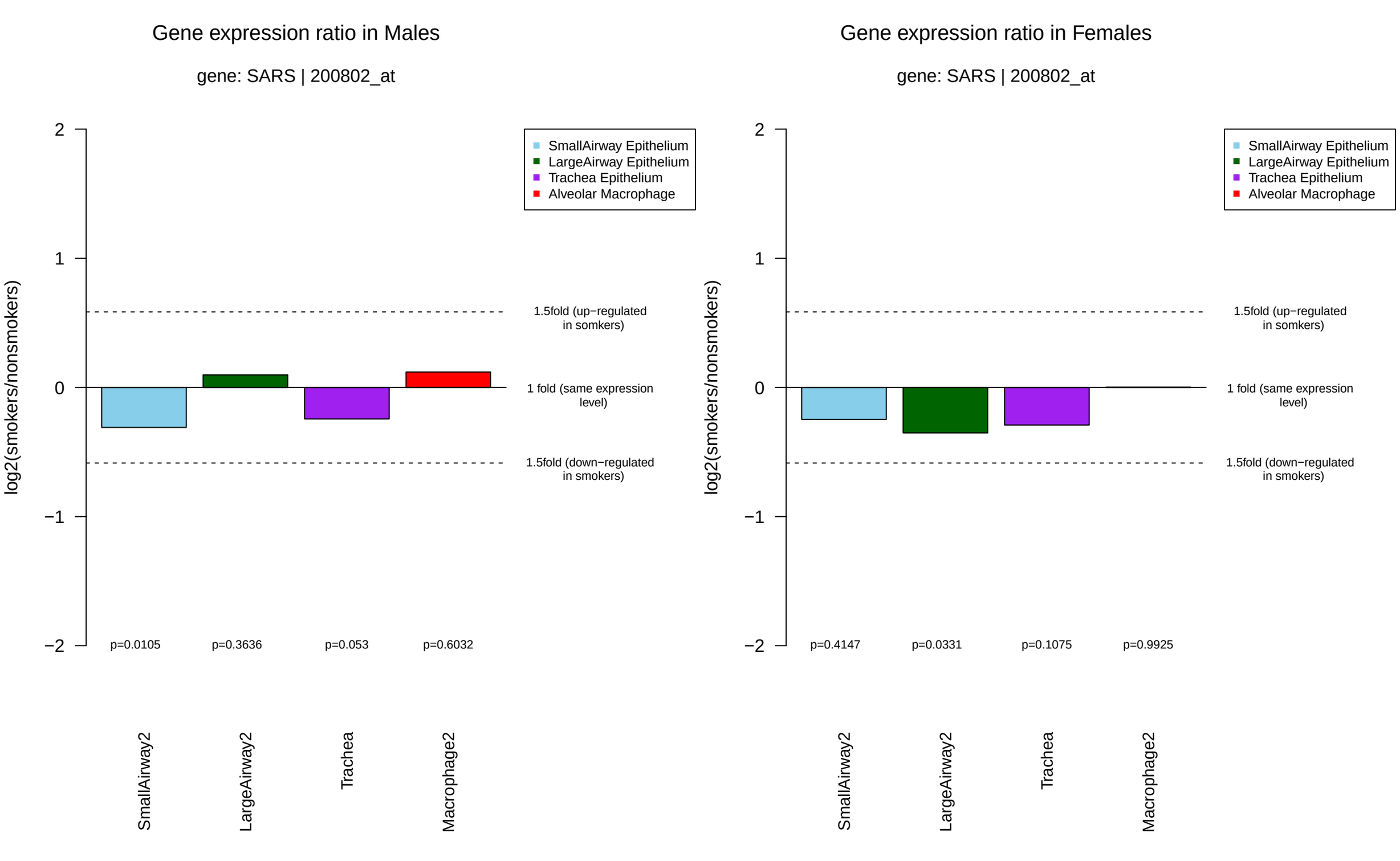
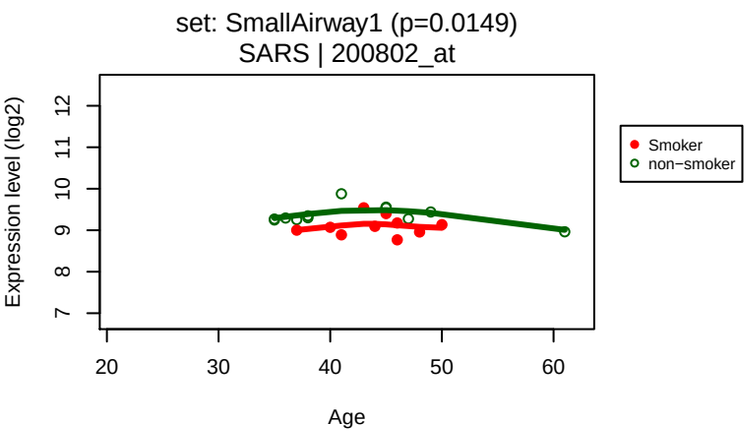
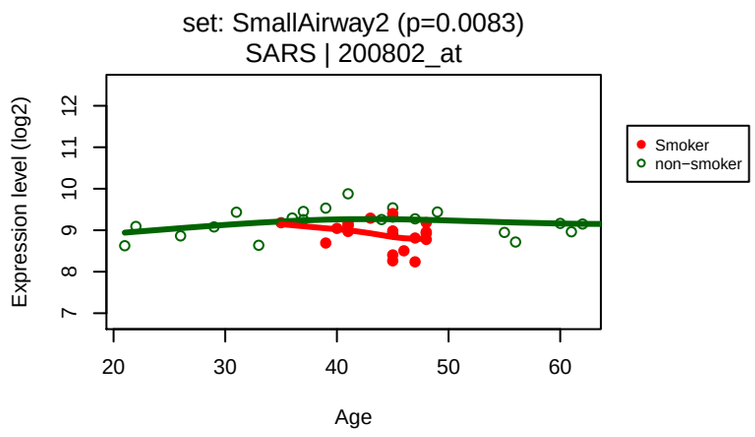
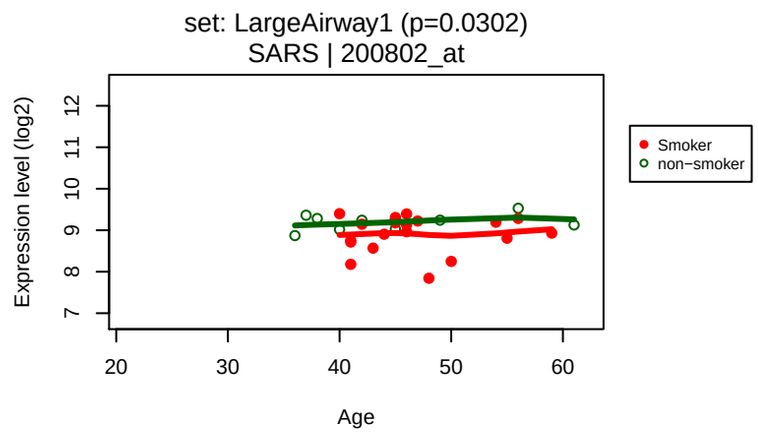

Gene expression of smokers and nonsmokers
gene: SARS | 200802_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0149
p=0.0083
p=0.0039
p=0.0302
p=0.5645
p=0.0267
p=0.289
p=0.634
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

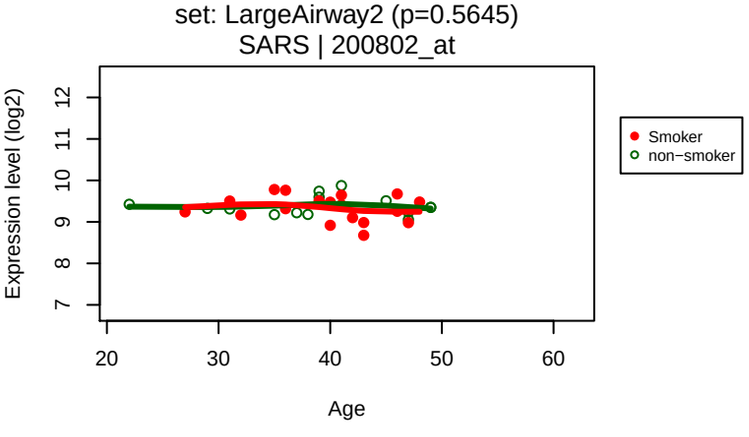

Gene expression ratio (smokers/nonsmokers)
gene: SARS | 200802_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0149
p=0.0083
p=0.0039
p=0.0302
p=0.5645
p=0.0267
p=0.289
p=0.634
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

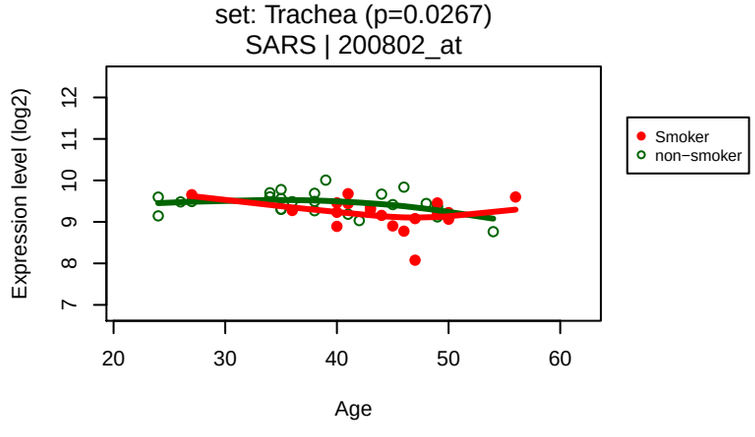

Gene expression ratio in Males
gene: SARS | 200802_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0105
p=0.3636
p=0.053
p=0.6032
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SARS | 200802_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4147
p=0.0331
p=0.1075
p=0.9925
SmallAirway2
LargeAirway2
Trachea
Macrophage2

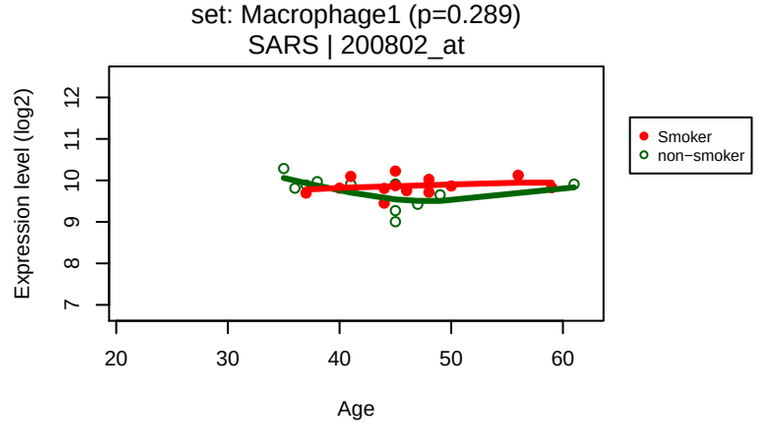

set: SmallAirway1 (p=0.0149)
SARS | 200802_at
Expression level (log2)
Age
Smoker
non−smoker

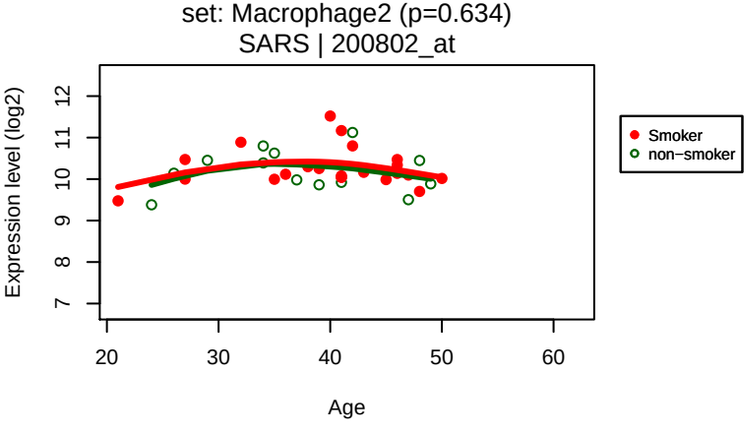

set: SmallAirway2 (p=0.0083)
SARS | 200802_at
Expression level (log2)
Age
Smoker
non−smoker

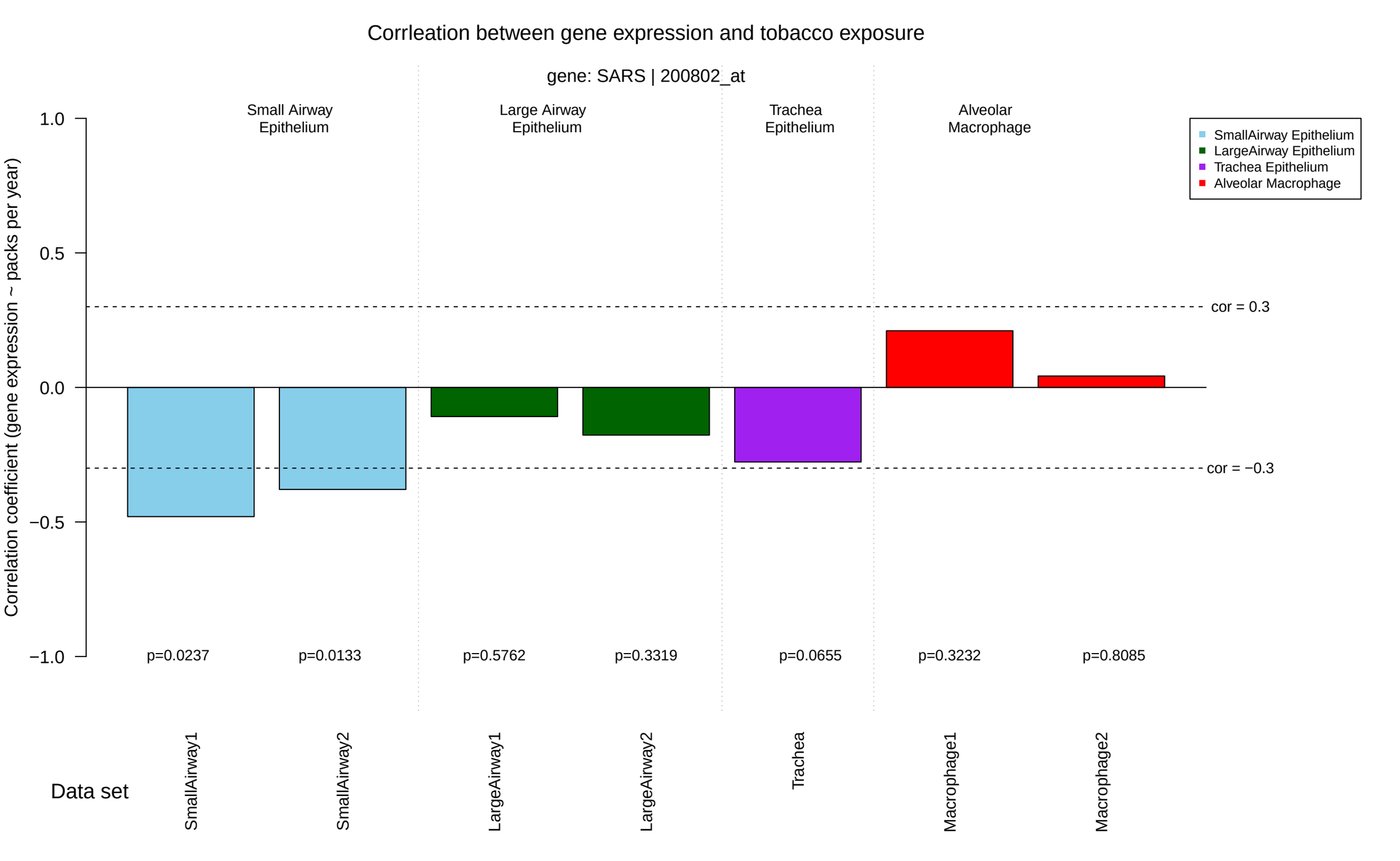

set: LargeAirway1 (p=0.0302)
SARS | 200802_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5645)
SARS | 200802_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0267)
SARS | 200802_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.289)
SARS | 200802_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.634)
SARS | 200802_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SARS | 200802_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0237
p=0.0133
p=0.5762
p=0.3319
p=0.0655
p=0.3232
p=0.8085
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2